

Review- Mathematical Epidemiology

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Abstract

This review is primarily for my own note-taking and understanding of the subject. Its main purpose is to give me a concise set of notes I can return to when required. As such, there are likely to be many mistakes, and it is a rather coarse-grained depiction of the subject.

Deterministic - SIS

This model uses two compartments; Susceptible and Infected. The former meaning an individual does not have the disease, but it is possible for them to have it, while the latter the individual currently has the disease. The model then switches between these two compartments or states. Such a model is generally applied to sexually transmitted diseases (STDs). For this model we make use of the following pair of ordinary differential equations (ODEs).

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta SI}{N} + (b + \gamma)I \\ \frac{dI}{dt} &= \frac{\beta SI}{N} - (b + \gamma)I\end{aligned}$$

Where S and I refer to the number of susceptible and infected individuals, $\beta > 0$ is the contact rate, $\gamma > 0$ is the recovery rate, $b \geq 0$ is the birth rate, and $N = S + I$ is total population. Generally we assume the birth rate equals the death rate so that N is constant. In this case the basic reproduction number is defined by:

$$R_0 = \frac{\beta}{b + \gamma}.$$

This model produces an analytic solution from which we get that:

1. If $R_0 \leq 1$, then $\lim_{t \rightarrow \infty} (S, I) = (N, 0)$. This is a disease-free equilibrium.
2. If $R_0 > 1$, then $\lim_{t \rightarrow \infty} (S, I) = (\frac{N}{R_0}, N(1 - \frac{1}{R_0}))$. This is an endemic equilibrium.

Further it is clear $\frac{dS}{dt} = -\frac{dI}{dt}$ The below figures demonstrate different possibilities for outcomes of this model.

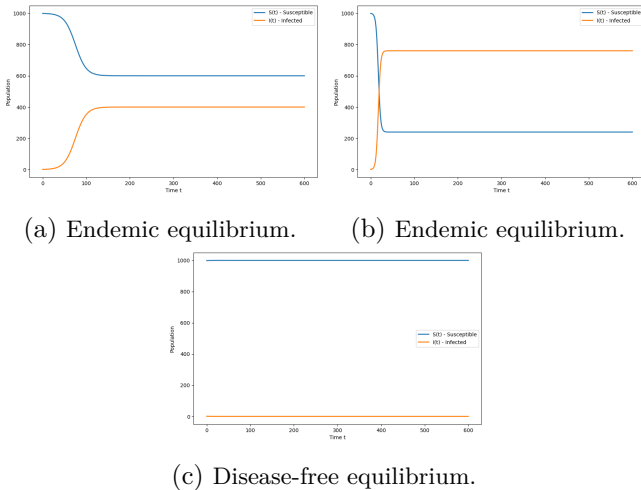


Figure 1:

Deterministic - SIR

The SIR model is an extension of the SIS model with 3 compartments Susceptible, Infected, and Recovered. This model is typically used to model disease such as Monekypox; where immunity is generally gained after recovering from the disease. This system is governed by the following ODEs:

$$\frac{dS}{dt} = -\frac{\beta SI}{N} + b(I + R)$$

$$\begin{aligned}\frac{dI}{dt} &= \frac{\beta SI}{N} - (b + \gamma)I \\ \frac{dR}{dt} &= \gamma I - bR\end{aligned}$$

Where we make use of the same notation as the SIS model. The ODEs here can be expressed in Logistic form in this case there is no analytic closed form solution, but we still have the following result;

1. If $R_0 \leq 1$, then $\lim_{t \rightarrow \infty} I(t) = 0$. We have a disease-free equilibrium.
2. If $R_0 > 1$, then $\lim_{t \rightarrow \infty} (S(t), I(t), R(t)) = (\frac{N}{R_0}, \frac{bN}{b + \gamma}(1 - \frac{1}{R_0}), (\frac{\gamma N}{b + \gamma}(1 - \frac{1}{R_0})))$. This is an endemic equilibrium.
3. Assume $b = 0$. If $R_0 \frac{S(0)}{N} > 1$, then there is an initial increase in the number of infected cases $I(t)$ (epidemic), but if $R_0 \frac{S(0)}{N} \leq 1$, then $I(t)$ decreases monotonically to 0 (disease-free equilibrium).

The below figures demonstrate some different possible outcomes for this model.

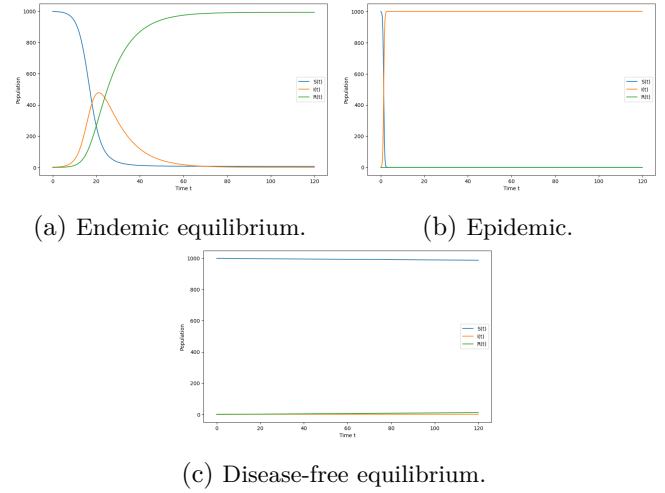


Figure 2:

DTMC - SIS

The discrete time Markov chain analogy to the dynamical SIS is really just a better approximation of what the deterministic version is trying to do. We model the states Susceptible and Infected, as states in a discrete time chain and we assume that the timestep Δt is sufficiently small that populations only increase or decrease by 1 in a given time step. The Markov property in this context is given by

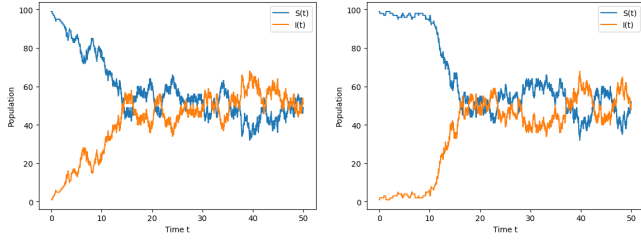
$$P\{I(t + \Delta t) \mid I(0), I(\Delta t), \dots, I(t)\} = P\{I(t + \Delta t) \mid I(t)\}$$

. We further assume the Markov chain is homogenous. Here we have an independent random variable, $I(t)$, from such a chain we develop a $(N+1) \times (N+1)$ tridiagonal transition probability matrix

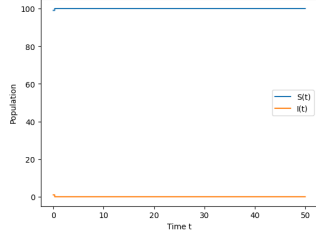
$$P = \begin{pmatrix} 1 & d(1)\Delta t & \dots & 0 \\ 0 & 1 - (b + d)(1)\Delta t & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 - d(N)\Delta t \end{pmatrix}$$

Out of laziness I skip most of the detail in this matrix. We have here $b(i) = \frac{\beta i(N-i)}{N}$ corresponds to an increase in the number of infections, $d(i) = (b + \gamma)i$ corresponds to a decrease, and $1 - d(i) - b(i)$ a recurrent step. Such movements form the tridiagonal elements of our transition matrix. In this case it can be shown

that the expectation for the number of infections is less than the deterministic solution. We further note that state 0 is the only recurrent state in this system, so that eventually we always end up in a diseases-free equilibrium and never an endemic equilibrium. The following plots show some possible outputs for this system.



(a) Slow absorption. (b) Slow absorption.



(c) Rapid absorption.

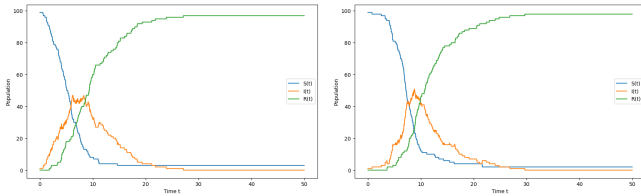
Figure 3:

DTMC - SIR

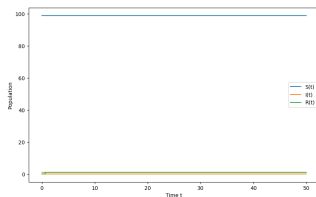
When we extend this system to the SIR model we now have two independent random variables and a state is characterised by the ordered pair $(S(t), I(t))$. In such a case we use the following transition probabilities:

$$p_{(s+k, i+j), (s, i)} = \begin{cases} \beta is / N \Delta t & \text{if } (k, j) = (-1, 1) \\ \gamma i \Delta t & \text{if } (k, j) = (0, -1) \\ bi \Delta t & \text{if } (k, j) = (1, -1) \\ b(N - i - s) \Delta t & \text{if } (k, j) = (1, 0) \\ 1 - \beta is / N b(N - i - s) \Delta t - (\gamma i + b(N - s)) \Delta t & \text{if } (k, j) = (0, 0) \\ 0 & \text{other} \end{cases}$$

From these transitions we have the Markov chain existing as a digraph on a two-dimensional lattice with $(N, 0)$ again as an absorbing state. The below figures demonstrate some possibilities for such a model.



(a) Slow absorption. (b) Slow absorption.



(c) Rapid absorption.

Figure 4:

CTMC - SIS

For the continuous time Markov chain models we have the states as discrete random variables, but time as a continuous quantity.

In such a case the Markov property is expressed as:

$$P\{I(t_{n+1}) | I(t_0), I(t_1), \dots, I(t_n)\} = P\{I(t_{n+1}) | I(t_0)\}$$

. For any sequence of non-negative real numbers $0 \leq t_0 < \dots < t_{n+1}$. In the CTMC case we need to consider infinitesimal Δt , as such we include a $\lambda(\Delta t)$. We cannot in this case write out the probability transition matrix in the same manner as we did before, we must instead make use of the Kolmogorov forward equation as state:

$$\frac{dP(t)}{dt} = P(t)Q.$$

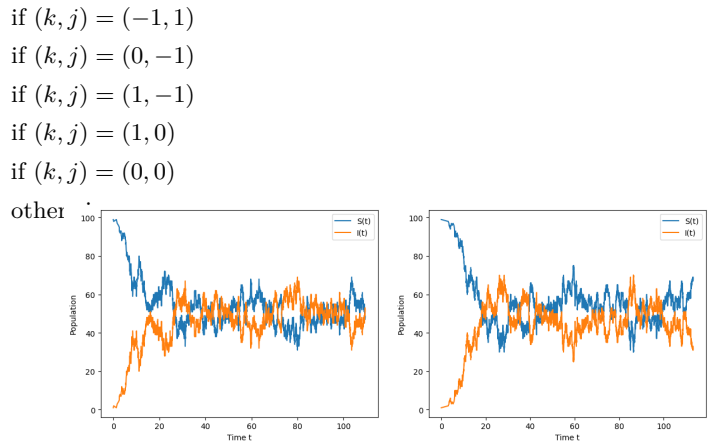
Where $P(t)$ is the probability transition matrix and Q is the generator or jump matrix. We further have:

$$Q = \lim_{t \rightarrow 0} \frac{P(t) - I}{t}$$

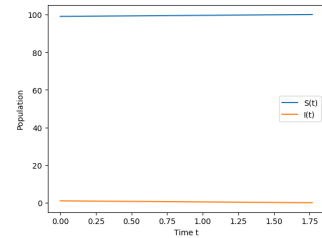
For the SIS model we produce a generator matrix:

$$Q = \begin{pmatrix} 0 & d(1) & \dots & 0 \\ 0 & -(b+d)(1) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -d(N) \end{pmatrix}$$

. We have maintained the same notation as for previous SIS models. In this case we have a similar result where absorption in the state 0, must occur. The below figures demonstrate some possible sample paths for this model we note that it is really just a more detailed version of the prior two SIS models in the sense that we allow time to run with continuity.



(a) Slow absorption. (b) Slow absorption.



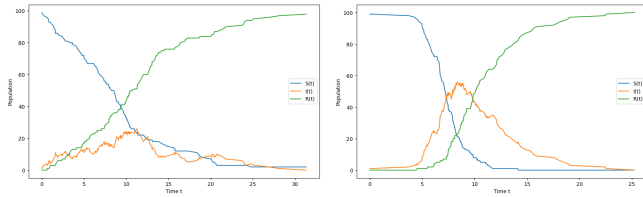
(c) Rapid absorption.

Figure 5:

CTMC - SIR

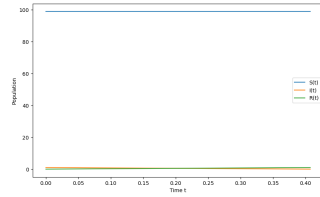
The CTMC SIR model follows rather as one would expect. A key point that should have been mentioned in the previous section is

that the inter event time must be exponentially distributed, this is due to the exponential distribution being unique in that it has no memory allowing for our required Markov property. The main difference in the SIR case is that the distribution is bivariate in $S(t)$ and $I(t)$. Nonetheless we can perform similar operations with the Kolmogorov forward equations to arrive at a similar result with some of the viable sample paths shown below.



(a) Slow absorption.

(b) Slow absorption.



(c) Rapid absorption.

Figure 6:

Beyond

The next step it seems clear is to introduce models on the basis of stochastic differential equations (SDEs), where we move to Wiener processes, that is what could be called a continuous time continuous state Markov chain. We will not report on such models here, but the next foundational additional work makes a full exploration of SDE analogies to work; the properties of such models and the properties of the models discussed here.